

3.2 Ionic bonding homework — answers

CIE 9701 AS Chemistry · 7 multiple-choice + 10 written marks

Do not issue this file to students.

Section A — Multiple choice

Quick answers

Q	Ans.	Q	Ans.	Q	Ans.
A1	B	A4	C	A7	C
A2	C	A5	D		
A3	B	A6	B		

Worked explanations

A1. B

2021 MJ Paper 13 Q6

Ionic bonding requires the *transfer* of electrons from a metal atom to a non-metal atom — two *different* elements. An element contains only one type of atom, so it cannot produce oppositely charged ions; ionic bonding is therefore *never* found in elements.

A: covalent — H_2 , Cl_2 , diamond (giant covalent).

C: metallic — Na, Fe, Cu.

D: van der Waals' forces *are* found in elements — between I_2 molecules or between noble-gas atoms.

A2. C

2025 ON Paper 11 Q16

The radii identify the elements. A metal loses electrons, so its ionic radius is *smaller* than its atomic radius; a non-metal gains electrons, so its ionic radius is *larger*. X: $0.118 \rightarrow 0.053$ nm — a metal cation, matches Al (Al 0.118 , Al^{3+} 0.054). Y: $0.099 \rightarrow 0.180$ nm — a non-metal anion, matches Cl (Cl 0.099 , Cl^- 0.181). Z: $0.160 \rightarrow 0.072$ nm — a metal cation, matches Mg (Mg 0.160 , Mg^{2+} 0.072). So X = Al, Y = Cl and Z = Mg.

A: X is aluminium, a metal that conducts electricity; Y is chlorine, a non-metal that does not. X has the *higher* conductivity, not lower.

B: Y is chlorine (a simple molecular gas, mp -101°C); X is aluminium (mp 660°C). Y has the *much lower* melting point.

C (correct): Y (Cl) and Z (Mg) are a non-metal and a metal, so they form the ionic compound MgCl_2 .

D: Z is magnesium, a metal — it forms ions by *losing* electrons, not gaining them.

A3. B

2025 MJ Paper 11 Q8

The key on the paper defines the symbols: **Na** and **Cl** stand for the *atoms*, while **+** and **−** stand for the *ions* (Na^+ and Cl^-). A sodium chloride crystal is a giant ionic lattice of *ions* in a regular alternating arrangement, so the correct diagram is the one built from **+** and **−** symbols (option B).

A: labels the particles Na and Cl, i.e. *atoms* (see the key). A sodium chloride lattice contains Na^+ and Cl^- ions, not atoms.

C / D: also use ion symbols, but the ions are not arranged in the correct regular alternating (rock-salt) lattice of sodium chloride — e.g. the packing and relative positions of the two ions are wrong.

A4. C

2023 MJ Paper 11 Q6

In the rock-salt structure each Na^+ ion is surrounded by **6** Cl^- ions (octahedral), and each Cl^- ion is surrounded by **6** Na^+ ions. The coordination numbers are therefore 6 and 6.

A / B: give 4 for one of the ions. A coordination number of 4 would be a different structure (e.g. zinc blende), not NaCl.

D: gives 8 for the sodium ions. 8:8 is the caesium chloride structure; Cs^+ is much larger than Na^+ so CsCl packs differently.

A5. D

2025 ON Paper 13 Q18

P_4O_{10} is a simple molecular oxide: discrete P_4O_{10} molecules held together by weak van der Waals' forces, so it has a simple structure. MgO , Al_2O_3 and SiO_2 all have *giant* structures (giant ionic for MgO and Al_2O_3 , giant covalent/macromolecular for SiO_2).

A: MgO is a giant ionic lattice — high melting point, conducts when molten.

B: Al_2O_3 is a giant ionic lattice (with some covalent character).

C: SiO_2 is a giant covalent (macromolecular) solid — this is the common trap, because it looks like a simple oxide of a non-metal but is macromolecular.

A6. B

2025 MJ Paper 14 Q12

Sodium ethanoate, CH_3COONa , contains *ionic* bonding (between the Na^+ ion and the CH_3COO^- ion) and *covalent* bonding (the C–H, C–C, C–O and C=O bonds inside the ethanoate ion). Neither of these is a coordinate (dative covalent) bond, so B is the answer.

A: Al_2Cl_6 contains covalent bonds *and* coordinate bonds (the $\text{Cl} \rightarrow \text{Al}$ dative bonds in the dimer), so it is excluded.

C: MgCl_2 is ionic only — it has no covalent bonding.

D: NH_4Cl contains ionic bonding plus covalent N–H bonds *and* a coordinate bond in the NH_4^+ ion (the bond formed when the lone pair on N donates to H^+), so it is excluded.

A7. C

2021 ON Paper 12 Q35

(1) **False** — SiO_2 is a giant covalent solid (mp about 1710°C) while SiCl_4 is simple molecular (mp -68°C); SiO_2 has the much higher melting point. (2) **True** — SiCl_4 is hydrolysed by water ($\text{SiCl}_4 + 2\text{H}_2\text{O} \longrightarrow \text{SiO}_2 + 4\text{HCl}$); NaCl simply dissolves in water without reaction. (3) **True** — SiCl_4 is simple molecular (weak van der Waals' forces) so its melting point is far lower than that of NaCl, a giant ionic lattice. Correct statements: 2 and 3 $\rightarrow$ C.

A / B: both include statement 1, which is false.

D: includes only statement 1 (false) and drops the two true statements.

Section B — Mark scheme clips

B1(a) [2]

Question	Answer	Marks
1(a)	M1 ionic = oppositely charged ions / cations AND anions	1
	M2 covalent = shared pair(s) of electrons AND two nuclei	1

B2(a)(i) [1]

2(a)(i)	Li ⁺ AND S ²⁻	1
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B2(a)(ii) [2]

2(a)(ii)	M1 giant	1
	M2 (many) strong force(s) of attraction between oppositely charged ions OR (many) strong ionic bond(s)	1

B3(a)(i) [1]

Question	Answer	Marks
1(a)(i)	sodium – metallic AND sodium oxide – ionic	1

B3(a)(iii) [1]

1(a)(iii)	(sodium and oxide) ions move (through the liquid)	1
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B4(a)(i) [1]

Question	Answer	Marks
3(a)(i)	NaCl AND MgCl ₂	1

B5(c)(ii) [2]

2(c)(ii)	- false - Al₂O₃ melting point is lower than MgO (but) charge on Al ion is greater than that on Mg	2
	Any 2 points = 1 mark , all 3 points = 2 marks	